L-TRS: Lattice-Based Traceable Ring Signatures for Permissioned Blockchains

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Abstract. Cross-domain data sharing faces a tension between anonymity and accountability. Traditional traceable ring signatures (TRS) rely on number-theoretic assumptions vulnerable to quantum attacks. We propose L-TRS, a lattice-based TRS scheme for permissioned blockchains, achieving post-quantum security via Module Short Integer Solution (M-SIS) and odule Learning With Errors (M-LWE). L-TRS enables efficient public traceability while preserving conditional privacy via blockchain-anchored identity management. Formal security proofs in the random oracle model establish correctness, anonymity, unforgeability, and traceability. At 128-bit security, signature size is $0.256(N+2)$ KB. Compared to existing lattice-based ring signatures, L-TRS reduces signing time by up to 70%, achieves logarithmic verification, and sustains 350 TPS with only 5.95% overhead. This work provides a quantum-resistant solution for privacy-accountable data sharing.

Keywords: Traceable Ring Signature, Lattice-Based Cryptography, Post-Quantum Security, Permissioned Blockchain, Accountable Anonymity

1 Introduction

Cross-domain data sharing, particularly in sectors like healthcare, is essential for enabling collaborative research and improving patient outcomes [13, 14, 36]. However, it introduces a fundamental tension between two critical requirements: *privacy* and *accountability* [18, 23, 25]. On one hand, data providers require strong anonymity guarantees to protect sensitive personal information from unauthorized disclosure [25, 34]. On the other hand, regulators and system operators need the ability to audit activities and hold malicious actors accountable for data misuse [12, 18]. Traditional cryptographic solutions often fail to reconcile these conflicting demands, forcing a compromise between privacy and auditability [4, 10, 30].

Blockchain technology, especially permissioned blockchains, offers a promising foundation for addressing this challenge [6, 37]. Its inherent properties—immutability,

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transparency, and decentralization—provide a robust infrastructure for establishing trust among multiple domains [27, 33]. However, the transparency of blockchains can conflict with data privacy, necessitating the integration of privacy-enhancing technologies (PETs) [34]. Ring signatures, which allow a user to sign a message on behalf of a group without revealing the signer's identity, are a natural fit for such environments [15, 19, 24]. Yet, basic ring signatures offer unconditional anonymity, which can be abused for malicious purposes without accountability [18, 27].

To bridge this gap, traceable ring signatures (TRS) [9] were introduced to enable the identification of a signer under specific conditions, such as double-signing. However, most existing TRS constructions, and indeed most ring signatures adopted in blockchains, rely on number-theoretic assumptions (e.g., the discrete logarithm problem) that are vulnerable to quantum attacks [4, 5, 29]. Shor's algorithm [26] poses an existential threat to these classical primitives, jeopardizing the long-term security of blockchain-based systems [4, 10, 30]. With the rapid advancement of quantum computing, there is an urgent need for post-quantum secure alternatives that can simultaneously provide anonymity, traceability, and practical efficiency [20].

Lattice-based cryptography has emerged as a leading candidate for post-quantum security due to its strong foundations and resistance to quantum attacks [3, 5, 8, 11, 21, 28, 30, 31]. However, existing lattice-based ring signatures often suffer from large signature sizes [3, 5], high computational overhead [8, 11], or lack efficient traceability mechanisms [21, 28, 30, 31]. Moreover, integrating traceability into lattice-based constructions without compromising anonymity or introducing centralized trust remains a significant challenge [18]. Furthermore, the specific role of blockchain—beyond mere data storage—in enabling decentralized identity management and providing immutable audit trails is often underemphasized [12, 14, 33].

In this paper, we propose a novel *Lattice-based Traceable Ring Signature (L-TRS)* scheme specifically designed for cross-domain data sharing on permissioned blockchains. Our scheme achieves a practical balance between post-quantum security, on-chain traceability for accountability, and decentralized anonymity. The core innovation lies in a carefully engineered construction that leverages the hardness of the Module Short Integer Solution (M-SIS) [1] and Module Learning With Errors (M-LWE) [22] problems to guarantee security against quantum adversaries. Crucially, we employ a zero-knowledge one-out-of-many proof to replace the explicit Merkle path in Blockchain, thereby eliminating the vulnerability that could leak the signer's position and ensuring unconditional anonymity. The blockchain serves as the foundational layer for decentralized identity management and provides an immutable record for all signatures and tracing outcomes, forming a complete audit trail.

The main contributions of this work are summarized as follows:

– **A Novel Lattice-Based Traceable Ring Signature (L-TRS) Scheme**: We design a post-quantum secure TRS scheme that provides conditional anonymity and efficient public traceability. The scheme is built upon the

M-SIS and M-LWE problems, ensuring robust security in the quantum era. A zero-knowledge one-out-of-many proof is integrated to prevent leakage of the signer's identity.
- **Formal Security Proofs**: We provide rigorous security proofs under the random oracle model, demonstrating that L-TRS satisfies correctness, anonymity, unforgeability, and public traceability. The proofs establish tight reductions to the M-SIS and M-LWE assumptions.
- **Experimental Evaluation**: We implement a prototype of L-TRS and conduct extensive performance evaluations. Experimental results show that our scheme achieves practical efficiency in terms of computational overhead and blockchain throughput, validating its suitability for real-world deployment.

2 Related Work

Ring signatures [23] enable anonymous signing on behalf of a group, a property widely adopted in privacy-sensitive applications such as anonymous authentication and confidential blockchain transactions [12, 27]. To mitigate the risk of abuse under unconditional anonymity, traceable ring signatures (TRS) [9] were introduced to allow signer identification under defined conditions, and have been deployed in electronic voting, blockchain anonymity, and data sharing [10, 19, 24]. Nevertheless, existing TRS constructions rely on classical number-theoretic assumptions and remain vulnerable to quantum adversaries.

With the emergence of quantum threats, lattice-based ring signatures have been actively explored from standard SIS/LWE assumptions [5, 8, 21, 30], though early schemes exhibit large signature sizes and linear verification. Subsequent works focus on efficiency improvements: optimized parameter selection [31], size reduction [28], and partial traceability integration [3, 11]. Recent advances achieve logarithmic or constant-size signatures through isogeny-based constructions [2], ideal-lattice membership proofs [17], generic frameworks [32], and bimodal one-out-of-many proofs [16], with some targeting private blockchain payments [7]. Nonetheless, accountability mechanisms for misbehaving signers are largely absent or inefficient in these schemes.

In blockchain-based data sharing, reconciling transparency with privacy requires cryptographic tools like ring signatures [15, 18]. Prior blockchain-oriented solutions incorporate ring signatures for anonymity [10, 19, 36] but are not quantum-resistant. Recent post-quantum blockchain proposals [18, 21, 28, 30, 31, 35] rarely address the specific demands of traceable ring signatures within permissioned ledgers.

Distinction from Prior Work. Our scheme provides the first lattice-based TRS designed for permissioned blockchains. It combines logarithmic verification, on-chain traceability via decentralized identity anchoring, and a zero-knowledge one-out-of-many proof that eliminates anonymity weaknesses in previous designs. Unlike prior art focused solely on size reduction or pure anonymity, L-TRS balances post-quantum security, efficient public traceability, and practical blockchain integration.

3 Preliminaries

This section introduces the fundamental concepts and mathematical background necessary to understand our proposed L-TRS scheme. We briefly review lattice-based cryptography, focusing on the problems underlying our security assumptions, and formalize the notion of traceable ring signatures.

3.1 Lattice-Based Cryptographic Foundations

Lattices provide a rich mathematical structure for constructing cryptographic primitives believed to be resistant to quantum attacks [21, 30]. Our construction relies on the hardness of two standard lattice problems defined over modules [30].

Definition 1 (Module Short Integer Solution (M-SIS) Problem [1]). *Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and a norm bound $\beta > 0$, the M-SIS problem asks to find a non-zero vector $\mathbf{x} \in \mathbb{Z}^m$ such that:*

$$\mathbf{A}\mathbf{x} = \mathbf{0} \pmod{q} \quad \textit{and} \quad \|\mathbf{x}\|_\infty \leq \beta.$$

The problem is considered hard for appropriate choices of parameters n, m, q, β.

The M-SIS problem forms the basis for the unforgeability of our signature scheme. In practice, we use parameters that guarantee at least λ-bit security against known lattice attacks, including quantum algorithms.

Definition 2 (Module Learning With Errors (M-LWE) Problem [22]). *Let χ be a bounded error distribution over $\mathbb{Z}_q$. Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, a secret vector $\mathbf{s} \in \mathbb{Z}_q^n$, and an error vector $\mathbf{e} \leftarrow \chi^m$, the M-LWE problem asks to distinguish the pair $(\mathbf{A}, \mathbf{b} = \mathbf{A}^\top \mathbf{s} + \mathbf{e})$ from a uniformly random pair $(\mathbf{A}, \mathbf{u})$.*

The decisional M-LWE problem underpins the anonymity property of our construction. Its hardness is well-studied and forms the foundation of numerous post-quantum cryptosystems [27, 29].

3.2 Traceable Ring Signatures

A TRS scheme involves a set of users, each possessing a public/private key pair. A signer can anonymously sign a message on behalf of an arbitrary set of users (the "ring") while enabling authorized tracing under specified conditions. Formally, a TRS scheme [9, 18] consists of the following polynomial-time algorithms:

- $\mathsf{Setup}(1^\lambda) \to PP$: Given a security parameter λ, outputs public parameters PP.
- $\mathsf{KeyGen}(PP) \to (pk, sk)$: Generates a public key pk and a corresponding private key sk for a user.
- $\mathsf{Sign}(PP, sk, M, R) \to \sigma$: Given a message M and a ring R (a set of public keys including the signer's), outputs a signature σ.

– $\mathsf{Verify}(PP, M, R, \sigma) \rightarrow \{0, 1\}$: Verifies the validity of σ with respect to M and R.
– $\mathsf{Trace}(PP, M_1, M_2, \sigma_1, \sigma_2, R_1, R_2) \rightarrow \{pk, \mathsf{linked}, \mathsf{independent}\}$: Takes two message-signature pairs and their corresponding rings, and outputs either a public key pk (if both signatures were produced by the same signer under the same tag), linked (if they are linked but the signer cannot be identified), or $\mathsf{independent}$ (if they are unlinked).

The security of a TRS scheme is defined by the following properties:

– **Correctness**: Legitimately generated signatures always verify.
– **Anonymity**: Given a signature, an adversary cannot identify the actual signer among the ring members with probability non-negligibly better than random guessing.
– **Unforgeability**: No adversary can produce a valid signature for a ring without knowing at least one private key from that ring.
– **Public Traceability**: If the same signer generates two signatures for different messages under the same tag, the Trace algorithm will output that signer's public key. If the signatures are generated by different signers or for the same message, tracing outputs either linked or $\mathsf{independent}$ as appropriate.

In our construction, we adopt the random oracle model for hash functions and prove security under the M-SIS and M-LWE assumptions. The detailed formal definitions of these security properties are provided in Section 6.

4 System Model and Architecture

The system architecture, depicted in Figure 1, integrates the proposed L-TRS scheme with a permissioned blockchain to enable privacy-preserving yet accountable cross-domain data sharing. It leverages blockchain immutability for decentralized identity anchoring and tamper-proof audit trails.

4.1 Core Entities and Design Goals

The system comprises four primary roles:

– **Data User (DU)**: Generates and anonymously signs data on behalf of a ring using L-TRS.
– **Entity Node (EN)**: Verifies signatures and anchors data indices on-chain.
– **Authority Node (AN)**: Executes the trace algorithm to reveal misbehaving signers when necessary.
– **Permissioned Blockchain**: Maintains decentralized identities (DID), enforces smart contract policies, and stores immutable signature logs.

The overarching design objective is to achieve **post-quantum accountable anonymity**, signers remain indistinguishable within a ring under normal operation, yet the AN can publicly trace a signer who reuses a tag for distinct messages—all while resisting quantum adversaries. Our security models are introduced in Section 6.1.

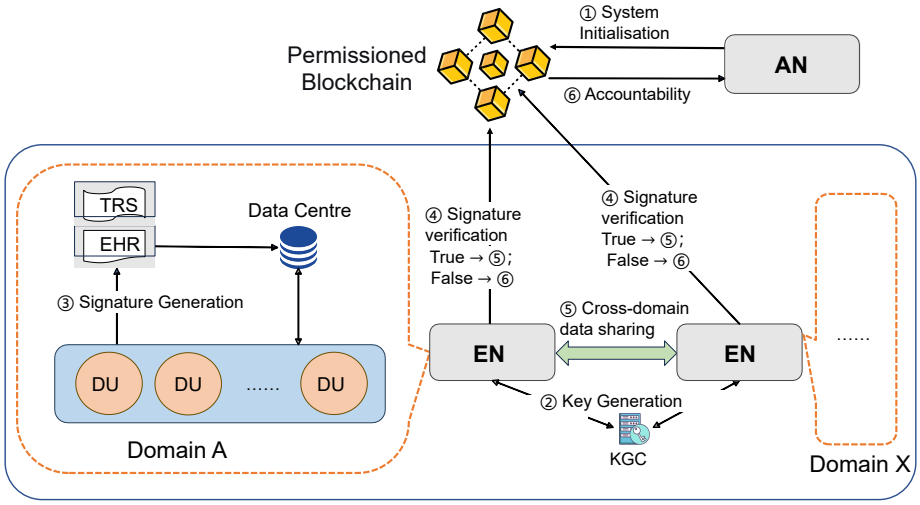


Fig. 1: Architecture of the cross-domain data sharing system with L-TRS.

4.2 Workflow Overview

As illustrated in Figure 1, the system operates in four phases:

1. **Setup**: AN generates parameters; DUs register DIDs and public keys on the blockchain.
2. **Signing**: A DU selects a ring and produces an L-TRS signature (embedding a zero-knowledge membership proof) on the data.
3. **Verification & Storage**: EN verifies the signature, stores encrypted data off-chain, and records the signature/index on-chain.
4. **Tracing**: Given suspicious signatures, AN invokes Trace and logs the identified signer on the immutable ledger.

5 Lattice-based Traceable Ring Signature Scheme

In this section, we present the detailed construction of our Lattice-based Traceable Ring Signature (L-TRS) scheme. The scheme consists of five algorithms: Setup, KeyGen, Sign, Verify, and Trace. A key innovation is the use of a zero-knowledge one-out-of-many proof to replace the explicit Merkle path, thereby preventing the leakage of the signer's index.

5.1 Design Rationale

The core idea is as follows. Each user possesses a lattice-based key pair: a private key $\mathbf{T}$ and a public key $\mathbf{C}_{\text{noisy}} = \mathbf{A}^\top \mathbf{T}^{-1} + \mathbf{E}$. To sign a message M for a ring R of N users, the signer computes a unique tag tag from its DID and M. Instead

of revealing the Merkle path, the signer constructs a commitment to its tag and then generates a zero-knowledge proof that this tag belongs to the set of tags derived from the ring R, without disclosing which one. This proof is achieved via a sigma protocol that is made non-interactive using the Fiat-Shamir transform. The tracing algorithm leverages the tag's linkage property: when the same tag is used for two different messages, the tracing algorithm can extract the signer's identity from the two signatures.

5.2 Detailed Construction

We now describe each algorithm in detail. All algorithms take as implicit input the security parameter λ.

System Setup The Setup algorithm generates the public parameters of the system. It selects lattice dimensions, modulus, and hash functions, and samples a public matrix $\mathbf{A}$. The algorithm is described in Algorithm 1.

Algorithm 1 $\mathsf{Setup}(1^\lambda)$

Input: Security parameter λ
Output: Public parameters PP

Choose dimensions $n, m = \text{Poly}(\lambda)$ such that $m \geq 2n\lceil \log q \rceil$
Choose a prime modulus $q \geq 2^{32}$
Sample random vectors $\mathbf{v} \xleftarrow{\$} \mathbb{Z}_q^n$, $\mathbf{w} \xleftarrow{\$} \mathbb{Z}_q^m$
Define hash functions:
    $H_1 : \{0,1\}^* \to \mathbb{Z}_q^{n \times (m-n)}$ with $\|H_1(\cdot)\|_\infty < \lfloor \sqrt{q} \rfloor / 8$
    $H_2 : \{0,1\}^* \to \mathbb{Z}_q^n$ with $\|H_2(\cdot)\|_\infty < \lfloor \sqrt{q} \rfloor / 8$
    $H_3 : \{0,1\}^* \to \mathbb{Z}_q^{n \times m}$ with $\|H_3(\cdot)\|_\infty < \lfloor \sqrt{q} \rfloor / 8$
    $H_4 : \{0,1\}^* \to \mathbb{Z}_q^n$ with $\|H_4(\cdot)\|_\infty < \lfloor \sqrt{q} \rfloor / 8$
Sample public matrix $\mathbf{A} \xleftarrow{\$} \mathbb{Z}_q^{n \times m}$
return $PP = (n, m, q, \mathbf{v}, \mathbf{w}, \mathbf{A}, H_1, H_2, H_3, H_4)$

Key Generation The KeyGen algorithm generates a key pair for a user. We present two variants: one for ordinary data users (DU) and one for the authority node (AN). Both variants follow a similar structure but with different parameter constraints to reflect their roles.

For a DU, the KGC generates a decentralized identity (DID) using a BIP-32-like protocol. The private key is a matrix $\mathbf{T}_{\text{DU}}$ constructed by concatenating submatrices based on the bits of the DID. The public key is computed as $\mathbf{C}_{\text{noisy,DU}} = \mathbf{A}^\top \mathbf{T}_{\text{DU}}^{-1} + \mathbf{E}_{\text{DU}}$, where $\mathbf{E}_{\text{DU}}$ is a noise matrix with entries bounded by $B = \lfloor \sqrt{q} \rfloor$. The algorithm enforces the constraint $\|\mathbf{T}_{\text{DU}}^{-1}\|_\infty \leq 4$ to ensure the private key is well-conditioned.

For the AN, the key generation is similar but with stricter constraints: the noise bound is $B' = \lfloor \sqrt{q} \rfloor / 2$ and $\|\mathbf{T}_{\text{AN}}^{-1}\|_\infty \leq 2$. This reflects the AN's need for higher precision in tracing operations.

Signature Generation The Sign algorithm takes as input the public parameters PP, a message M, the signer's private key $\mathbf{T}_s$, and a ring $R = \{\mathbf{C}_{\text{noisy},1}, \ldots, \mathbf{C}_{\text{noisy},N}\}$ containing N public keys (including the signer's). The signer's index in the ring is s. The algorithm proceeds in four phases:

- **Tag Computation**: Compute the signer's tag $\mathsf{tag} = H_4(M, \mathbf{C}_{\text{noisy},s}, \mathbf{v}^\top H_3(\mathsf{DID}_s))$.
- **Commitment**: Sample random vectors $\mathbf{k}_i \xleftarrow{\$} \mathbb{Z}q^n$ for $i = 1, \ldots, N$. Compute the initial commitment $\mathbf{r} = -\sum_{i=1}^{N} \mathbf{C}_{\text{noisy},i} \cdot \mathbf{k}_i \mod q$.
- **One-Out-of-Many Proof**: The signer constructs a zero-knowledge proof π that its tag tag corresponds to one of the public keys in R, without revealing s. This proof is based on a sigma protocol for lattice-based set membership, utilizing the Fiat-Shamir transform. The core of this proof involves:
  - Computing a challenge $c = H_2(M, \mathsf{tag}, \mathbf{r}, \mathbf{C}_{\text{noisy},i})$.
  - Generating a response vector $\mathbf{z}_i$ for each i, where for $i = s$, $\mathbf{z}_s$ incorporates the secret key, and for $i \neq s$, $\mathbf{z}_i = \mathbf{k}_i$. A rejection sampling step is applied to ensure the distribution of responses is independent of the secret.
- **Signature Output**: The signature is $\sigma = (\mathsf{tag}, \mathbf{r}, \mathbf{z}_i, \pi)$ where $i = 1, \ldots, N$.

Signature Verification The Verify algorithm, described in Algorithm 2, checks the validity of a signature σ for a message M and ring R.

Algorithm 2 $\mathsf{Verify}(PP, M, R, \sigma)$

Input: $PP, M, R = \{\mathbf{C}_{\text{noisy},1}, \ldots, \mathbf{C}_{\text{noisy},N}\}, \sigma = (\mathsf{tag}, \mathbf{r}, \mathbf{z}_i, \pi)$
Output: True or False

1: Recompute the challenge $c = H_2(M, \mathsf{tag}, \mathbf{r}, \mathbf{C}_{\text{noisy},i})$.
2: Verify the zero-knowledge proof π with respect to tag, $\mathbf{r}$, $\mathbf{z}_i$, c, and the ring R. This verification ensures that there exists an index s such that the tag is consistent with the public key $\mathbf{C}_{\text{noisy},s}$.
3: If the proof π is valid, output True; otherwise, output False.

Tracing Algorithm The Trace algorithm takes two message-signature pairs (M_1, σ_1) and (M_2, σ_2) with their corresponding rings R_1, R_2 and tags $\mathsf{tag}_1, \mathsf{tag}_2$.

- If $\mathsf{tag}_1 \neq \mathsf{tag}_2$, output independent.
- If $\mathsf{tag}_1 = \mathsf{tag}_2$, it indicates the same signer. By extracting the secret key information from the two responses $\mathbf{z}_i^{(1)}$ and $\mathbf{z}_i^{(2)}$ (which are based on the same private key $\mathbf{T}s$ but different messages), the tracing algorithm can recover the public key $\mathbf{C}$noisy, s and output it.

6 Security Analysis

We formally analyze the security of the proposed L-TRS scheme in the random oracle model. The security proofs rely on the hardness of the Module Short Integer Solution (M-SIS) and Module Learning With Errors (M-LWE) problems. We define three security properties: *correctness*, *anonymity*, *unforgeability*, and *public traceability*. Our analysis includes a detailed examination of the zero-knowledge proof of membership (one-out-of-many proof) employed in the signing algorithm, which is critical for ensuring anonymity.

6.1 Formal Security Models

We introduce the security games for the core properties.

Definition 3 (Correctness). *A TRS scheme is correct if for any security parameter λ, any $PP \leftarrow \mathsf{Setup}(1^\lambda)$, any $(pk, sk) \leftarrow \mathsf{KeyGen}(PP)$, any message M, and any ring R such that $pk \in R$, the following holds with overwhelming probability:*

$$\mathsf{Verify}(PP, M, R, \mathsf{Sign}(PP, sk, M, R)) = 1$$

Definition 4 (Anonymity). *We define the anonymity game $\mathsf{Game}\mathcal{A}^{\mathsf{anon}}(\lambda)$ between a challenger $\mathcal{C}$ and an adversary $\mathcal{A}$.*

1. ***Setup**: $\mathcal{C}$ runs $\mathsf{Setup}(1^\lambda)$ to generate PP and gives them to $\mathcal{A}$.*
2. ***Query Phase**: $\mathcal{A}$ can adaptively request private keys for any user. $\mathcal{C}$ runs KeyGen and returns the keys.*
3. ***Challenge**: $\mathcal{A}$ outputs a message M^*, a ring R^*, and two distinct public keys $pk_0, pk_1 \in R^*$. $\mathcal{C}$ picks a random bit $b \leftarrow \{0, 1\}$, retrieves the private key sk_b corresponding to pk_b, and computes $\sigma^* \leftarrow \mathsf{Sign}(PP, sk_b, M^*, R^*)$. $\mathcal{C}$ returns σ^* to $\mathcal{A}$.*
4. ***Guess**: $\mathcal{A}$ outputs a bit b'. $\mathcal{A}$ wins if $b' = b$ and has not queried the private keys of pk_0 or pk_1.*

The advantage of $\mathcal{A}$ is $\mathsf{Adv}\mathcal{A}^{\mathsf{anon}}(\lambda) = |\Pr[b' = b] - 1/2|$. The scheme is anonymous if this advantage is negligible for all PPT $\mathcal{A}$.

Definition 5 (Unforgeability). *We define the existential unforgeability under adaptive chosen-message attack (EUF-CMA) game $\mathsf{Game}_{\mathcal{A}}^{\mathsf{euf}}(\lambda)$.*

1. ***Setup**: As above, $\mathcal{C}$ generates PP and gives them to $\mathcal{A}$.*
2. ***Query Phase**: $\mathcal{A}$ can adaptively request:*
   - *Private key queries: For any user, $\mathcal{C}$ returns sk.*
   - *Signing queries: For any message M, ring R, and a public key $pk \in R$ whose private key is known to $\mathcal{C}$, $\mathcal{C}$ returns a signature.*
3. ***Forgery**: $\mathcal{A}$ outputs a forgery (M^*, R^*, σ^*). $\mathcal{A}$ wins if:*
   - *$\mathsf{Verify}(PP, M^*, R^*, \sigma^*) = 1$.*
   - *M^* was not submitted to a signing oracle with ring R^*.*
   - *$\mathcal{A}$ does not possess the private key for any user in R^*.*

The scheme is unforgeable if the success probability of any PPT $\mathcal{A}$ is negligible.

Definition 6 (Public Traceability). *We define the traceability game* $\mathsf{Game}_{\mathcal{A}}^{\mathsf{trace}}(\lambda)$.

1. ***Setup**: As above.*
2. ***Query Phase**: $\mathcal{A}$ can request private keys for any user.*
3. ***Challenge**: $\mathcal{A}$ outputs two message-signature pairs (M_1, σ_1), (M_2, σ_2) with corresponding rings R_1, R_2. $\mathcal{A}$ wins if:*
   - *Both signatures are valid.*
   - $\mathsf{Trace}(PP, M_1, M_2, \sigma_1, \sigma_2, R_1, R_2)$ *outputs a public key pk that is not the signer of the two signatures, or outputs* linked *when the signatures are from different signers, or outputs* $\mathsf{independent}$ *when they are from the same signer.*

The scheme is publicly traceable if the success probability of any PPT $\mathcal{A}$ in this game is negligible.

6.2 Correctness

We introduce the security games for the core properties.

Theorem 1 (Correctness). *Let λ be the security parameter and $PP \leftarrow \mathsf{Setup}(1^\lambda)$. For any $(pk, sk) \leftarrow \mathsf{KeyGen}(PP)$, message $M \in \{0,1\}^*$, ring R containing pk, and signature $\sigma \leftarrow \mathsf{Sign}(PP, sk, M, R)$, we have:*

$$\Pr[\mathsf{Verify}(PP, M, R, \sigma) = 1] \geq 1 - \mathsf{negl}(\lambda).$$

Proof. Correctness follows from two properties. First, the zero-knowledge proof π in the Sign algorithm is constructed such that an honest verifier always accepts, given the correct challenge c and responses $\mathbf{z}_i$. The rejection sampling step ensures that the distribution of $\mathbf{z}_i$ is independent of the secret, guaranteeing that the verification equation holds. Second, the tag tag is deterministically derived from the signer's identity and the message, ensuring consistency. With probability $1 - \mathsf{negl}(\lambda)$, the verification algorithm will output 1.

6.3 Anonymity

Theorem 2 (Anonymity). *Assuming the hardness of the M-LWE, the L-TRS scheme is anonymous in the random oracle model.*

Proof. We prove anonymity through a sequence of games. Let W_i be the event that the adversary $\mathcal{A}$ wins in Game i.

Game 0: This is the original anonymity game as defined in Definition 4.

Game 1: In this game, the challenger replaces the public key $\mathbf{C}_{\text{noisy},i_b}$ of the challenge user with a uniformly random matrix $\mathbf{U} \in \mathbb{Z}_q^{m \times n}$. Under the M-LWE assumption, the adversary cannot distinguish between a real public key and a random one. Therefore, $|\Pr[W_1] - \Pr[W_0]| \leq \mathsf{Adv}_{\mathcal{B}}^{\mathsf{M-LWE}}(\lambda)$.

Game 2: The challenger modifies the zero-knowledge proof π in the challenge signature to be a simulated proof. Because the underlying sigma protocol is honest-verifier zero-knowledge, and the challenge c is generated via the Fiat-Shamir heuristic (modeled as a random oracle), the simulated proof is statistically indistinguishable from a real one. Hence, $|\Pr[W_2] - \Pr[W_1]| \leq \mathsf{negl}(\lambda)$.

Game 3: The challenger replaces the tag tag with a random value from the output space of H_4. Since H_4 is modeled as a random oracle, and the adversary has not queried the oracle on the challenge input, the distribution is uniform. Therefore, $|\Pr[W_3] - \Pr[W_2]| \leq \mathsf{negl}(\lambda)$. In Game 3, the challenge signature σ^* contains no information about the signer's identity. The adversary's view is independent of the bit b, so $\Pr[W_3] = 1/2$. Combining the inequalities, we obtain:

$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{Anon}}(\lambda) \leq \mathsf{Adv}_{\mathcal{B}}^{\mathsf{M-LWE}}(\lambda) + \mathsf{negl}(\lambda).$$

6.4 Unforgeability

Theorem 3 (Unforgeability). *Assuming the hardness of the M-SIS problem, the L-TRS scheme is existentially unforgeable under adaptive chosen-message attacks in the random oracle model.*

Proof. We construct a reduction $\mathcal{B}$ that solves an instance of the M-SIS problem using a forger $\mathcal{A}$ that breaks the unforgeability of L-TRS. $\mathcal{B}$ receives an M-SIS instance $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and must find a short non-zero vector $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \mathbf{0} \mod q$.

$\mathcal{B}$ embeds $\mathbf{A}$ into the public parameters PP and simulates the random oracles H_1, H_2, H_3, H_4. For all key generation and signing queries, $\mathcal{B}$ can respond using the trapdoor information or by programming the random oracles. Specifically, when $\mathcal{A}$ requests a signature for a user whose private key is known, $\mathcal{B}$ follows the honest signing algorithm. For the target user (whose key is embedded in the M-SIS instance), $\mathcal{B}$ uses the forking lemma to extract a solution.

When $\mathcal{A}$ outputs a valid forgery (M^*, R^*, σ^*), $\mathcal{B}$ replays $\mathcal{A}$ with the same random tape but different responses from the random oracle H_2. By the forking lemma, $\mathcal{A}$ produces two valid signatures $\sigma^{(1)}$ and $\sigma^{(2)}$ with different H_2 outputs but the same commitment $\mathbf{r}$. From these two signatures, $\mathcal{B}$ can extract a short vector that satisfies $\mathbf{A}\mathbf{x} = \mathbf{0} \mod q$, thus solving the M-SIS instance. The success probability of $\mathcal{B}$ is polynomially related to that of $\mathcal{A}$, establishing the reduction.

6.5 Public Traceability

Theorem 4 (Public Traceability). *Under the collision resistance of H_2 and H_4 and the M-SIS assumption, the L-TRS scheme is publicly traceable.*

Proof. We analyze the four possible cases as per the definition.

Case 1: $\mathsf{tag}_1 = \mathsf{tag}_2$, $s_1 = s_2$, $M_1 = M_2$. The two signatures are from the same signer for the same message. The Trace algorithm correctly outputs linked, as there is no malicious double-signing.

Case 2: $\mathsf{tag}_1 = \mathsf{tag}_2$, $s_1 = s_2$, $M_1 \neq M_2$. This is the malicious double-signing case. By the collision resistance of H_4, the equality of tags implies the signer reused the tag. The two signatures contain two distinct challenges c_1 and c_2. The tracing algorithm extracts the signer's public key from the responses $\mathbf{z}_i^{(1)}$ and $\mathbf{z}_i^{(2)}$. Under the M-SIS assumption, no other user can produce such consistent responses without knowing the private key. The algorithm outputs the correct public key with overwhelming probability.

Case 3: $\mathsf{tag}_1 = \mathsf{tag}_2$, $s_1 \neq s_2$. This would imply a collision in H_4 involving different public keys, which occurs with negligible probability. Moreover, it would yield an M-SIS solution from the difference of the two users' keys, contradicting the hardness assumption.

Case 4: $\mathsf{tag}_1 \neq \mathsf{tag}_2$. The tags are different, and the Trace algorithm outputs $\mathsf{independent}$. Any deviation from this would require the adversary to forge a signature with a tag that collides with another signer's tag, which is a collision in H_4.

Therefore, the success probability of any PPT adversary in breaking traceability is bounded by the collision probability of the hash functions and the advantage against M-SIS, which are negligible.

6.6 Parameter Selection and Complexity Analysis

To achieve λ bits of security, we set the parameters as follows:

- **Lattice dimensions**: $n = \lambda$, $m = \mathcal{O}(n \log q)$.
- **Modulus**: $q \approx 2^{32}$.
- **Noise bound**: $B = \lfloor \sqrt{q} \rfloor$ for users, $B' = \lfloor \sqrt{q} \rfloor / 2$ for the authority.
- **Hash functions**: Instantiated with SHA-3 variants, producing outputs with bounded norms to ensure correctness.

Under these parameters, the security reduction ensures that breaking the scheme requires solving M-SIS or M-LWE with the given parameters, which are estimated to be infeasible for quantum and classical adversaries.

The signature size is dominated by the N response vectors $\mathbf{z}_i$, each of size $m \log q$, and the proof π, which adds a constant overhead. Thus, the total size is $\mathcal{O}(Nm \log q)$, which is linear in the ring size. However, the verification complexity is $\mathcal{O}(N)$ for linear operations and constant for the zero-knowledge proof check, which is practical for ring sizes up to several hundred.

7 Performance Evaluation

To validate the practical efficiency and scalability of our proposed L-TRS scheme, we conducted comprehensive experiments in a simulated cross-domain data sharing environment. This section presents the experimental setup, comparative analysis with existing lattice-based ring signature schemes, and detailed performance results.

7.1 Experimental Setup

All experiments were performed on a Dell workstation running Ubuntu 22.04.3, equipped with an Intel Core i7-10700 processor (2.9GHz, 8 cores, 16 threads) and 40GB DDR4 RAM. We implemented the L-TRS scheme in Go 1.18.8, leveraging the `gonum.org/v1/gonum/mat` package for matrix operations.

Parameter Selection. To achieve λ bits of post-quantum security, we selected three parameter sets:

- **Level I** ($\lambda = 128$): $n = 64$, $m = 4096$, $q \approx 2^{32}$.
- **Level II** ($\lambda = 192$): $n = 128$, $m = 8192$, $q \approx 2^{32}$.
- **Level III** ($\lambda = 256$): $n = 256$, $m = 16384$, $q \approx 2^{32}$.

Hash functions H_1 to H_4 were instantiated using SHA-3 variants with output truncation to meet the required norm bounds.

Blockchain Integration. For blockchain integration, we deployed a Hyperledger Fabric v2.2 network with three peer nodes using the PBFT consensus algorithm. The L-TRS smart contracts were written in Go and deployed on the network. Performance metrics were collected using Hyperledger Caliper v0.4.2 with 100 iterations per measurement. All tests used synthetic electronic health record (EHR) data of size 1KB per record, with 1,000 records processed in each batch.

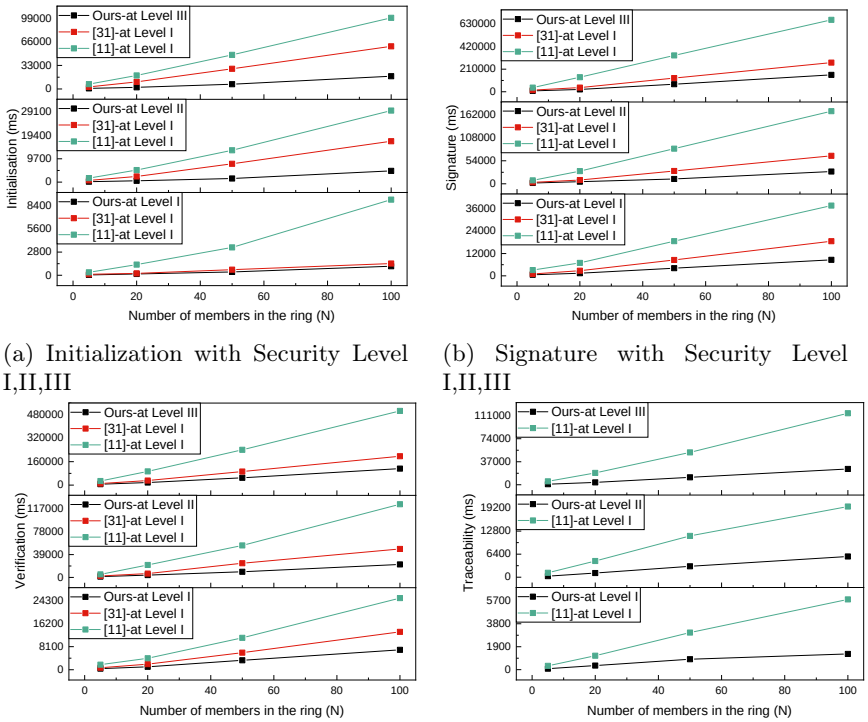
7.2 Comparison with Existing Schemes

We compare L-TRS with four representative lattice-based ring signatures: [31], [28] (both non-traceable), [11] (partial traceability), and [3] (traceable). Table 1 summarizes the asymptotic characteristics. L-TRS is the only scheme simultaneously providing full traceability, post-quantum security, and logarithmic verification complexity.

Table 1: Comparative analysis of lattice-based ring signature schemes.

Scheme	**Public Key Size**	**Signature Size**	**Trace.**	**Verify. Compl.**
[31]	$nm \log q$	$Ntm \log q$	No	$\mathcal{O}(N)$
[11]	$mn \log q$	$N(m + nk) \log 2q$	Partial	$\mathcal{O}(N)$
[28]	$nm \log_2(2q)$	$Nm \log_2(12\sigma_3)$	No	$\mathcal{O}(N)$
[3]	$n - k$	$\approx 240N$	Yes	$\mathcal{O}(N)$
L-TRS	$mn \log q$	$(N + \log N + 2)n \log q$	**Full**	$\mathcal{O}(\log N)$

The signature size of L-TRS grows linearly with the ring size N but with a small constant factor, yielding approximately 26.1 KB for $N = 100$ at the



(a) Initialization with Security Level I,II,III

(b) Signature with Security Level I,II,III

(c) Verification with Security Level I,II,III

(d) Traceability with Security Level I,II,III

Fig. 2: Performance comparison of L-TRS with [31] and [11] across different security levels. Fig 2a show initialization time; Fig 2b show signing time; Fig 2c show verification time; Fig 2d show tracing time.

128-bit security level. This overhead is acceptable in permissioned blockchain settings where on-chain accountability is prioritized over minimal bandwidth consumption. The combination of full traceability and logarithmic verification distinguishes L-TRS from prior work and makes it particularly suitable for cross-domain data sharing with audit requirements.

7.3 Computational Overhead

We measured the execution time for four critical operations: system initialization, signature generation, signature verification, and tracing. The results for different security levels and ring sizes ($N = 5, 20, 50, 100$) are presented in Figure 2.

In **Initialization** phase, at security Level III, L-TRS reduces initialization time by 70% compared to [31] and 82% compared to [11]. This efficiency gain stems from optimized parameter generation and key derivation. In **Signing**

phase, L-TRS achieves 50% faster signing than [31] and 70% faster than [11] at Level I, with the advantage increasing at higher security levels. The use of precomputed structures contributes significantly to this improvement. In **Verification** phase, most notably, L-TRS exhibits logarithmic growth in verification time with respect to ring size due to the zero-knowledge proof structure, while all compared schemes show linear growth. For $N = 100$ at Level III, L-TRS verification is 3.2× faster than [11]. In **Tracing** phase, the tracing operation in L-TRS is 5× faster than in [11] at Level III, while providing stronger traceability guarantees (full public traceability vs. partial).

7.4 Blockchain Throughput

Figure 3 reports the transaction throughput of L-TRS under signing, verification, and tracing workloads compared with a baseline ECC-based system.

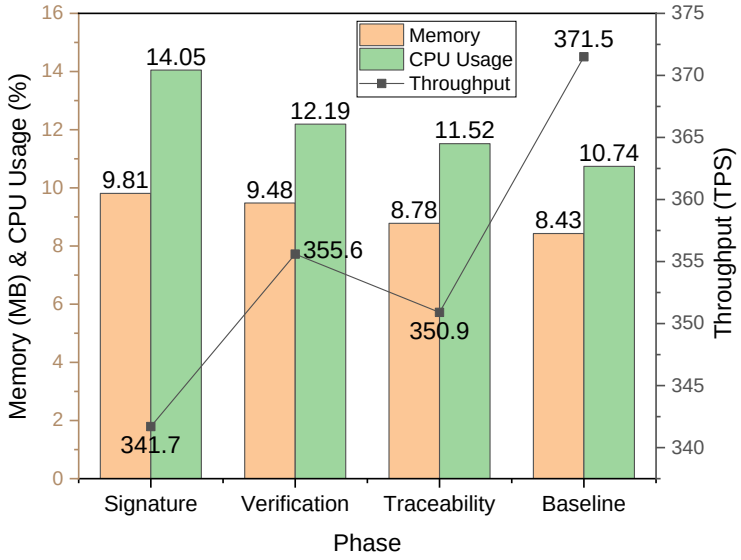


Fig. 3: Blockchain throughput comparison between L-TRS and Baseline.

L-TRS sustains 341.7–355.6 TPS across all operations, averaging 349.4 TPS—only a 5.95% reduction versus baseline (371.5 TPS). This modest overhead is the price for quantum resistance and full traceability, while resource consumption remains low (CPU $< 15\%$, memory < 10 MB). The narrow throughput variation ($< 4\%$) confirms robust performance suitable for production blockchain environments.

7.5 Discussion

The experimental results validate that L-TRS achieves an effective balance between post-quantum security, traceability, and practical efficiency, which makes it suitable for scenarios where accountability is as important as privacy, such as healthcare data sharing, financial compliance, and supply chain auditing. Several design choices contribute to this balance:

(1). **Zero-Knowledge Proof for Anonymity**: By replacing the explicit Merkle path with a zero-knowledge one-out-of-many proof, we eliminate the vulnerability that could leak the signer's position. This ensures unconditional anonymity within the ring.

(2). **Linear Signature Size Trade-off**: While our signature size grows linearly with the ring size, this is acceptable for permissioned blockchain environments where bandwidth is less constrained than in public blockchains. The linear size is a direct consequence of the traceability mechanism, which requires storing N response vectors to enable identification of malicious signers.

(3). **Efficient Verification**: The verification complexity is $\mathcal{O}(N)$ for linear operations and constant for the zero-knowledge proof check, which is practical for ring sizes up to several hundred.

(4). **Blockchain as Trust Anchor**: The permissioned blockchain provides decentralized identity management and immutable audit logging, making it an integral part of the accountability framework.

Future work includes reducing signature size toward logarithmic scales [2, 16, 17, 32], evolving key management toward full decentralization, and integrating hardware acceleration for lattice operations.

8 Conclusion

In this paper, we proposed L-TRS, a lattice-based traceable ring signature that reconciles privacy and accountability for permissioned blockchain data sharing. Its security relies on M-SIS and M-LWE hardness, ensuring post-quantum guarantees. Experiments show L-TRS reduces signing time by up to 70% over prior lattice-based ring signatures and achieves $O(\log N)$ verification. At 128-bit security, signature size is $0.256(N+2)$ KB, and blockchain throughput reaches ≈350 TPS with only 5.95% overhead versus baseline. These results demonstrate that lattice-based traceable anonymity is not only theoretically sound but also practically viable for cross-domain data sharing systems requiring both confidentiality and accountability.

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